UNIT 1

Introduction

Sessions 1 and 2

Three machines wait for you this year: one that detects a fire, one that shows live data on its own screen, and one that drives itself along a line. All three rest on the same foundation, and this unit builds it.

By the end of this unit you will be able to:

- Explain the Sense - Think - Act idea with your own example.
- Name every component in your kit and say what it does.
- Build a circuit on a breadboard and explain why it works.
- Tell a digital signal from an analog one and say which pins handle each.

SESSION 1 one class

Systems That Sense, and What Is in Your Box

Today you will

- Understand what makes something a system rather than a gadget.
- Learn the names of all sixteen components in your kit.
- Agree the rules of the robotics lab.

A robot is any machine that can **sense** something about the world, **decide** what that means, and then **act**. The automatic door at a shop, the smoke alarm on your ceiling and the reversing camera in a car are all robots by that definition, though none of them looks like one.

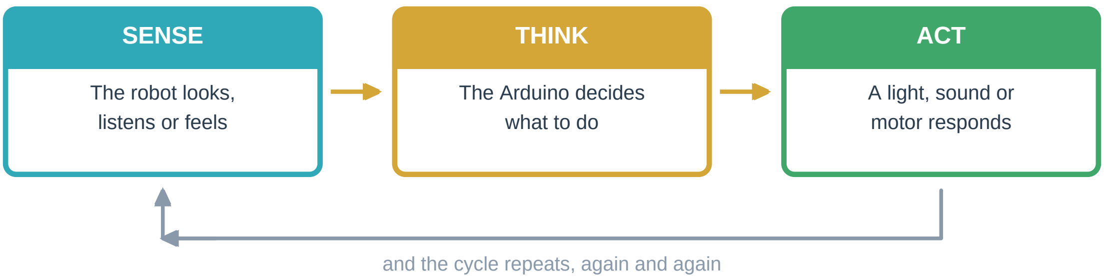


Figure 1.1 - Every robot in the world follows this same three-step loop

This year adds a word to that idea: **system**. A gadget does one thing. A system is several parts working together, where the behaviour of the whole is more than any single part. Your fire alarm will use two different sensors that must *agree* before it acts. Your robot will use two sensors whose readings change because of the robot's own movement. Thinking about the whole, not just the parts, is the real step up in Class 9.

Try This

> Think about a lift. List everything it senses (buttons, floor position, door obstruction, weight), everything it decides, and everything it acts on. Then find one case where two of its senses must agree before it does anything. Every safety system in the world is built on that idea.

1.1 Open the Box

Lay everything out before you plug anything in. Learn the names first - it is impossible to ask for help with 'that small blue thing'. Your kit is grouped by unit, so the colour of each label tells you roughly when you will first need that part.

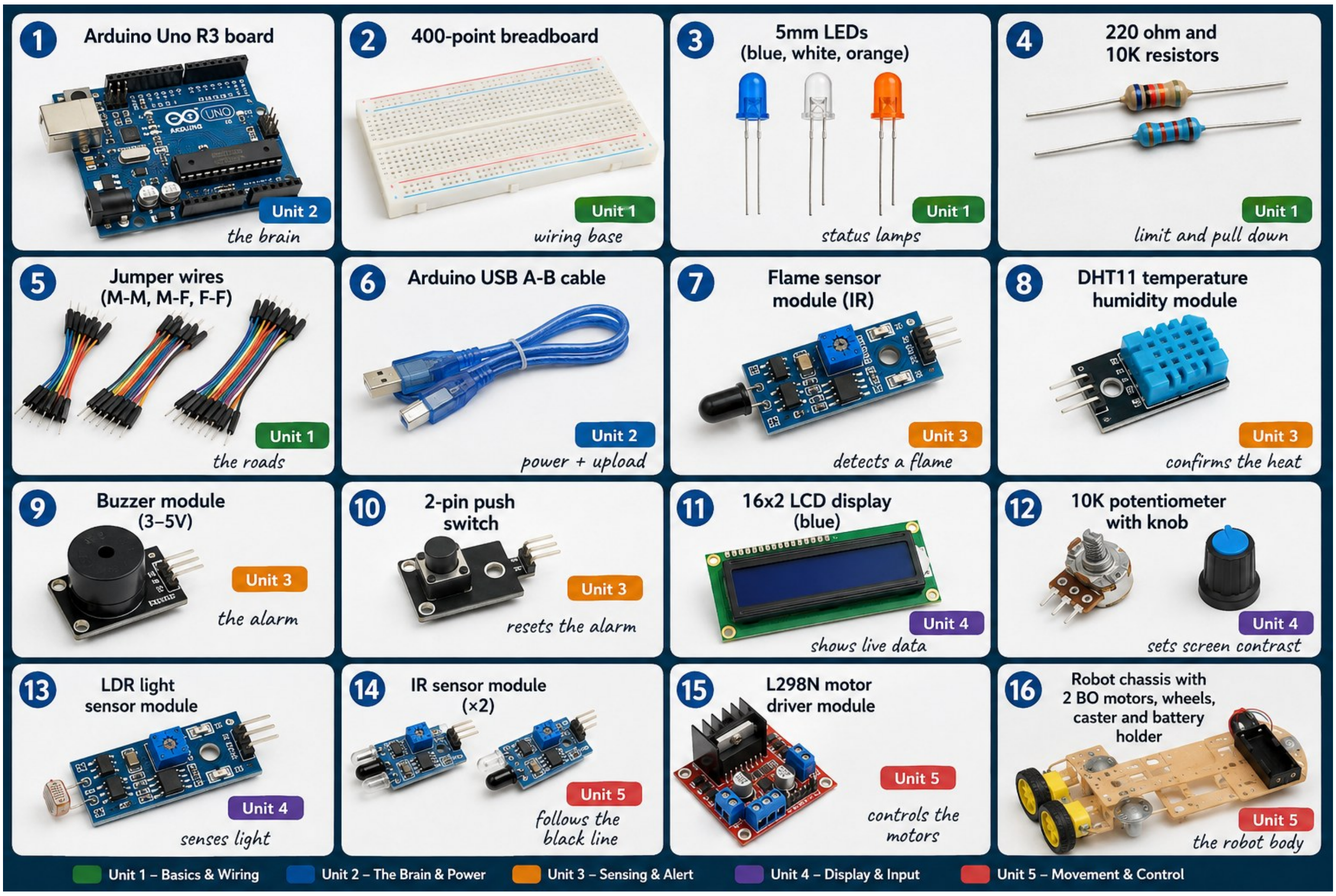


Figure 1.1 - Your kit. Learn these sixteen names before the next class.

Part	What it does	First used in
Arduino Uno board	The brain. It stores your program and runs it forever, even after the power has been off.	Session 3
USB A-B cable	Carries your program into the board and powers it while you test.	Session 3
400-point breadboard	Joins components together without any soldering.	Session 2
Jumper wires	The roads between parts. Male-to-male, male-to-female, female-to-female.	Session 2
5mm LEDs	Status lamps: normal, warning, alarm.	Session 2
220 ohm and 10K resistors	220 protects an LED; 10K pulls a switch input down to a known level.	Session 2
Flame sensor module	Detects the infrared light a fire gives off.	Session 6
DHT11 module	Measures temperature and humidity, to confirm what the flame sensor saw.	Session 7
Buzzer module	The alarm itself.	Session 7
2-pin push switch	Lets a human reset the alarm.	Session 8

Part	What it does	First used in
16x2 LCD display	Shows live readings with no laptop attached.	Session 9
10K potentiometer	Sets the contrast of the display.	Session 10
LDR light sensor module	Reports how bright the surroundings are.	Session 11
IR sensor modules (two)	Look down at the floor and tell black from white.	Session 12
L298N motor driver	A power amplifier for motors. The Arduino cannot drive them directly.	Session 12
Chassis with motors, wheels, caster and battery holder	The robot's body and its power supply.	Session 13

1.2 Rules of the Robotics Lab

1. Always unplug the USB cable before you change any wire.
2. Check the long leg of every LED and the + side of every module before power goes on.
3. Never run a wire straight from 5V to GND. That is a short circuit and it can kill the board.
4. **Never connect a motor directly to an Arduino pin.** It draws far more current than the pin can supply and the damage is permanent.
5. **Disconnect the battery pack before rewiring**, and never leave it connected at the end of a session.
6. **Flame testing only with a supervised candle in a metal tray**, away from paper, curtains and the battery pack.
7. Lift a robot off the floor before uploading, so it cannot drive away mid-upload.
8. Count every component back into the kit box before you leave.

> **Watch Out**
> Rule 6 is not a formality. You are building a fire detector, so at some point you must show it a real flame. One candle, one metal tray, one trainer watching, and nothing flammable within an arm's length. A fire alarm project that starts a fire is not a good demonstration.

SESSION 2

one class

Electricity, Circuits and Signals

Today you will

- Refresh how a circuit works and why it must be a loop.
- Light an LED on a breadboard with no code at all.
- Learn the difference between digital and analog signals.

Electricity behaves much like water in a pipe. The supply is the pump, the wires are the pipes, and each component is something the flow must pass through. **Voltage** is how hard the pump pushes, in volts. **Current** is how much flows each second, in amps. **Resistance** is anything that makes flowing harder, in ohms. These three are tied together by Ohm's law: **V = I x R**.

That law is not decoration - it tells you why a 220 ohm resistor is the right partner for an LED. An LED drops about 2 volts across itself, leaving 3 volts across the resistor. Current is then 3 divided by 220, which is about 0.0136 amps, or 13.6 milliamps. That is comfortably inside both what the LED can survive and what an Arduino pin can supply. Choose 22 ohms instead and the current would be ten times higher, and something would fail.

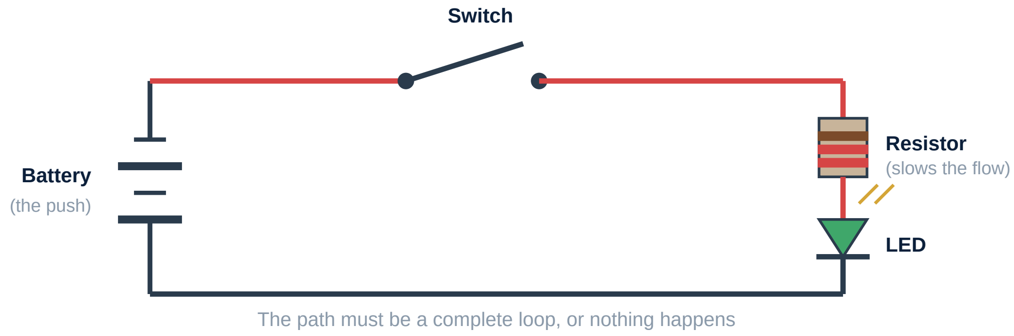
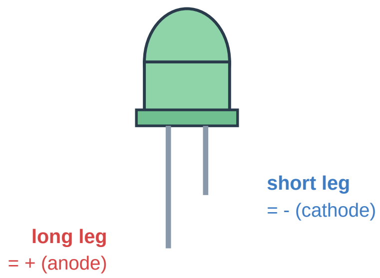


Figure 1.2 - A simple circuit: battery, switch, resistor and LED

LED

Light Emitting Diode



Put the long leg towards +. Backwards, it simply will not glow (and that is not a fault).

RESISTOR

220 ohm (red - red - brown - gold)

It has no + or - side.
Fit it either way round.

Its job is to protect the LED from too much current, like a narrow gate in a wide road.

> **Watch Out**
>
> An LED has a long leg and a short leg. Long leg towards +, short leg towards -. If an LED does not glow, turn it around before checking anything else. And it always needs a resistor: connect one straight to 5V and it will glow brilliantly for about a second, then never again.

2.1 Inside the Breadboard

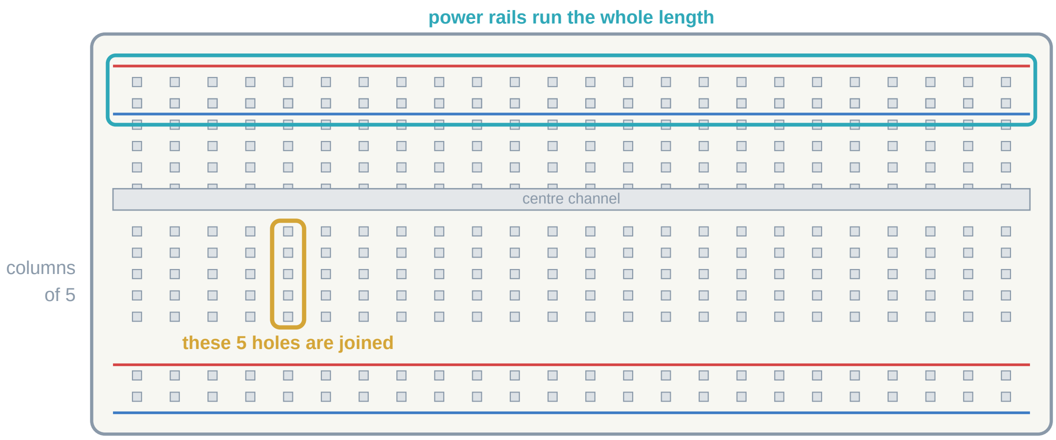


Figure 1.3 - Inside a breadboard: which holes are secretly connected

- The **long side rails** marked + and **-** run the whole length. Use them for power and ground.
- In the middle, holes are joined in **columns of five**, up and down only, never across.

- The **centre channel** splits the board in two. Nothing crosses it - which is exactly why chips are designed to straddle it.
- Two legs in the same column of five are connected; two legs in different columns are not.

> **Try This**
> Build the LED circuit with no code: LED legs in different columns, a 220 ohm resistor sharing the column with the long leg, resistor to 5V, short leg column to GND. Plug in the USB and it glows. Pull the GND wire and it dies. You have just proved the loop rule with your own hands.

2.2 Two Kinds of Signal

A push switch has two positions with nothing in between - a **digital** signal. A light level changes smoothly - an **analog** signal. The Arduino handles both: digital pins D0 to D13 deal in HIGH and LOW, while analog pins A0 to A5 measure a voltage and report a number from 0 to 1023.

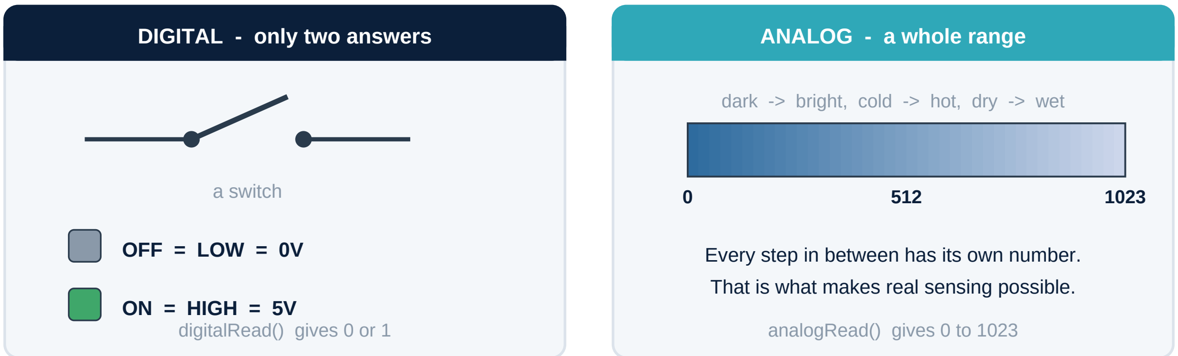


Figure 1.2 - The big idea of Class 7: numbers instead of yes-or-no

Your kit contains sensors of both kinds, and some that are cleverer than either. The LDR module gives you both: a raw analog value on AO and a simple yes-or-no on DO. The DHT11 is different again - it is a digital sensor that sends its readings as a coded sequence of pulses, which a library decodes for you. Knowing which kind you are holding decides which pin it goes to.

Sensor in your kit	Kind	Connect to
Push switch	Digital	any digital pin, with a 10K pull-down resistor
Flame sensor DO	Digital	any digital pin
Flame sensor AO	Analog	A0 to A5
LDR module AO	Analog	A0 to A5
Potentiometer wiper	Analog	A0 to A5
DHT11 DATA	Digital, coded	any digital pin, decoded by a library
IR sensor OUT	Digital	any digital pin

New word	What it means
Voltage	How hard electricity is pushed. Measured in volts.
Current	How much electricity flows each second. Measured in amps.
Ohm's law	$V = I \times R$. The relationship between voltage, current and resistance.
Pull-down resistor	A resistor that holds an input at 0V until a switch pulls it high.
System	Several parts working together, where the whole does more than any one part.